Data Flow Diagram

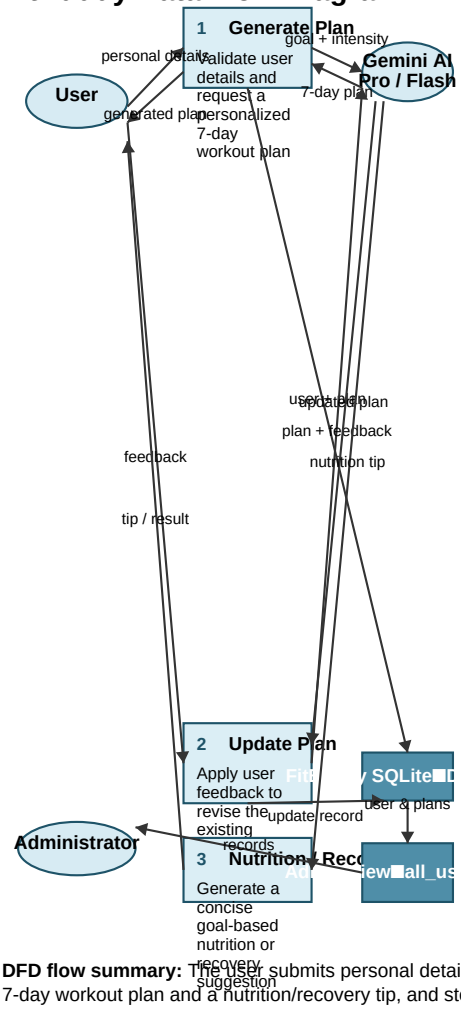
Date	01 October 2026
Team ID	SWTID-2026-6912
Project Name	FitBuddy - AI Fitness Plan Generator using Gemini Models
Team Leader	Pavithra M

Data Flow Diagram

The DFD below shows how user information moves through FitBuddy between the user, FastAPI processing modules, Gemini AI services, the SQLite database, and the administrator view.

Symbol	Name	Description
Oval / Rounded shape	External Entity	A person or system outside the application that sends or receives data (e.g., User, Administrator).
Rectangle with numbered header	Process	An activity that transforms incoming data into an output (e.g., Generate Plan, Update Plan).
Solid-fill rectangle	Data Store	A place where data is stored for later use (e.g., FitBuddy SQLite Database).
Labeled arrow	Data Flow	Movement of data between entities, processes, and data stores, labeled with the data being passed.

FitBuddy Data Flow Diagram



DFD flow summary: The user submits personal details and fitness preferences. FitBuddy validates the request, uses Gemini models to generate a 7-day workout plan and a nutrition/recovery tip, and stores user and plan information in SQLite. Feedback is sent back through the update process to revise the plan. The administrator can view stored users and original/updated plans through the admin view.

Team Work Assignment

Work is assigned in the same order as the names provided, aligned with the FitBuddy project workflow and milestones.

No.	Name	Role	Assigned Work
1	Pavithra M	Team Leader	Project coordination; model selection and architecture; Gemini integration oversight; milestone tracking; final review and submission.
2	Manisha S	Team Member	FastAPI backend and routing; user-input processing; routes.py development; integration of workout generation and feedback routes.
3	Monica G	Team Member	Frontend development; index.html, result.html and all_users.html; CSS styling; Jinja2 dynamic templates and responsive UI.
4	Monisha Anandan	Team Member	Database and storage; SQLAlchemy/SQLite setup; save/update/retrieve functions; user and workout-plan persistence.
5	Mridula M	Team Member	Testing and deployment; local setup and verification; endpoint testing; documentation support; final testing checklist and bug reporting.

Project Workflow Alignment

Milestone	Primary Responsibility
1. Model Selection & Architecture	Pavithra M
2. Core Functionalities & FastAPI Backend	Manisha S
3. routes.py Development	Manisha S
4. Frontend Development & Jinja2	Monica G
5. Database / Data Persistence	Monisha Anandan
6. Deployment, Testing & Verification	Mridula M
Final Review & Coordination	Pavithra M + all members

Note: The assignments are based on the project workflow described in the supplied FitBuddy project document, covering Gemini model selection, FastAPI/backend development, frontend/Jinja2, SQLite/SQLAlchemy storage, and deployment/testing.